A. Objective Type Questions

1. Which of the following best describes the relationship between Math and AI?

a) AI replaces the need for math

b) AI uses mathematical concepts to recognize patterns and make predictions

c) Math is not required for AI

d) AI is only about coding

Answer: b) AI uses mathematical concepts to recognize patterns and make predictions

2. Which branch of math is used to predict future events?

a) Algebra

b) Calculus

c) Probability

d) Geometry

Answer: c) Probability

3. Which of the following is a real-life application of Statistics in AI?

a) Calculating distance between two points

b) Predicting diseases in a population

c) Estimating slope of a line

d) Measuring angles of a triangle

Answer: b) Predicting diseases in a population

4. What is the heart of Artificial Intelligence?

a) Calculus

b) Linear Algebra

c) Statistics

d) Probability

Answer: b) Linear Algebra

5. In probability, if an event is certain to happen, its probability is:

a) 0

b) 1

c) ½

d) Undefined

Answer: b) 1

6. Which mathematical branch helps AI algorithms to improve through training?

a) Statistics

b) Calculus

c) Geometry

d) Trigonometry

Answer: b) Calculus

7. Which of the following events is impossible ?

a) Getting a head or tail when tossing a coin

b) Getting a 6 on a die

c) The sun rising in the west

d) Rain in monsoon

Answer: c) The sun rising in the west

2. Fill in the Blanks

1. Mathematics provides the _____ tools to develop AI algorithms.

Answer: mathematical

2. Math is the study of _____ and _____ in data.

Answer: patterns, relationships

3. AI algorithms can learn _____ just like humans.

Answer: patterns

4. Statistics is used for collecting, _____, and analyzing data.

Answer: exploring

5. Linear algebra helps to represent data in _____ and _____.

Answer: matrices, vectors

6. Probability tells how _____ something is to happen.

Answer: likely

3. True / False

1. Math is not related to AI. — False
2. Probability helps in making predictions in AI. — True
3. Calculus divides complex problems into simpler parts. — True
4. Statistics has no real-life use in AI. — False
5. Certain events always have a probability of 0. — False

4. Match the Following

Column A (Branch of Math)	Column B (Use in AI)
1. Statistics	a. Predicting events
2. Calculus	b. Representing data as matrices
3. Linear Algebra	c. Analyzing data

| 4. Probability | d. Training and improving models |

Answer:

1 → c, 2 → d, 3 → b, 4 → a

B. Subjective Questions

Short Answer

1. Why is mathematics essential for Artificial Intelligence?

Answer:

Mathematics provides the tools and logic needed to create AI algorithms. It helps in analyzing data, identifying patterns, and making accurate predictions in areas like healthcare, finance, and robotics.

2. How are Math and AI related?

Answer:

Math helps AI analyze and learn patterns from data. AI algorithms use mathematical models to solve problems, recognize images, and make predictions just like humans do.

3. Explain any two branches of math used in AI with examples.

Answer:

Statistics:Used to analyze and interpret data (e.g., predicting weather or disease outbreaks).

Probability: Used to calculate the likelihood of events (e.g., coin toss or traffic estimation).

4. What role does Calculus play in AI?

Answer:

Calculus helps AI models learn and improve by breaking complex systems into smaller parts, allowing algorithms to optimize performance and make better decisions.

5. How is Linear Algebra important in AI?

Answer:

Linear Algebra represents data as matrices and vectors, which are essential for data organization and processing in machine learning and neural networks.

Long Answer

1. Explain the uses of the four types of mathematics in AI with examples.

Answer:

Statistics: Helps collect and analyze data (e.g., predicting diseases).

Calculus: Trains AI models and improves accuracy (e.g., determining slope or area).

Linear Algebra: Structures data in matrices or vectors (e.g., image recognition).

Probability: Predicts outcomes (e.g., weather forecasting).

Together, these mathematical branches help AI learn, analyze, and predict intelligently.

2.Discuss any three real-life applications of Statistics in AI.

Answer:

Disaster Management: Predicts areas likely to face natural calamities.

Disease Prediction: Helps identify disease trends to plan vaccinations.

Weather Forecasting: Analyzes past data to predict future weather conditions.

3. What is Probability? Explain its types with real-life examples.

Answer:

Probability measures how likely an event is to occur.

Certain Events: The sun rises in the east.

Likely Events: It may rain in monsoon.

Unlikely Events: It may snow in Delhi.

Impossible Events: The sun rising in the west.

Equal Probability: Getting head or tail in a coin toss.

C. Assertion–Reason Type Questions

Directions: Each question consists of two statements – Assertion (A) and Reason (R).

Choose the correct option:

a) Both A and R are true, and R is the correct explanation of A.

b) Both A and R are true, but R is not the correct explanation of A.

c) A is true, but R is false.

d) A is false, but R is true.

---

1. Assertion (A): Mathematics is the foundation of Artificial Intelligence.

   Reason (R): AI uses mathematical formulas for problem-solving and prediction.

   Answer: a

2. Assertion (A): Calculus helps AI to train and improve its models.

   Reason (R): Calculus helps in representing data in rows and columns.

   Answer: c

3. Assertion (A): Linear Algebra is used to represent and process large data sets in AI.

Reason (R): It uses matrices and vectors for organizing data.

Answer: a

4. Assertion (A): Probability is used to find the exact result of an uncertain event.

Reason (R): Probability only tells how likely an event is to occur.

Answer: c

D. Scenario-Based Questions

1. Scenario 1:

During the rainy season, a weather app predicts there is a 75% chance of rain tomorrow .

Question:

Which branch of math is the app using?

What does the percentage represent?

Answer:

The app is using Probability .

The percentage represents the likelihood of the event (rain) occurring.

---

2. Scenario 2:

An AI model analyzes past sales data to determine which product is sold the most each month.

Question:

Which branch of math is being used?

What measure of central tendency helps in this analysis?

Answer:

Statistics

Mode

---

3. Scenario 3:

A self-driving car uses sensors to detect nearby vehicles and predict collisions.

Question:

Which mathematical concepts help the car identify distances and improve its model over time?

Answer:

Calculus and Linear Algebra

---

4. Scenario 4:

AI-based traffic control predicts traffic jams in certain areas during peak hours using past data.

Question:

Which branch of math is applied here and why?

Answer:

Probability and Statistics , because they analyze data patterns to predict future traffic density.

---

5. Scenario 5:

A healthcare AI system predicts the likelihood of a patient developing diabetes based on previous records.

Question:

Which mathematical branch supports this prediction?

Answer:

Statistics and Probability

PROBABILTY QUESTIONS:

A. Multiple Choice Questions (MCQs)

1. When a coin is tossed once, what is the probability of getting a head?

a) 0

b) ½

c) 1

d) 2

Answer: _____

2. A die is rolled. What is the probability of getting an even number?

a) ½

b) ⅓

c) ⅔

d) ¼

Answer: _____

3. What is the probability of getting a number greater than 4 on a die?

a) 1/3

b) 1/2

c) 1/6

d) 2/3

Answer: _____

4. If you pick a card at random from a deck of 52, what is the probability of getting a King?

a) 1/52

b) 1/26

c) 1/13

d) 4/52

Answer: _____

5. Which of the following has a probability of 0?

a) Getting head on tossing a coin

b) Getting 8 on a standard die

c) Getting tail on tossing a coin

d) Drawing a red card from a deck

Answer: _____

6. What is the probability of getting tails twice in two tosses of a fair coin?

a) ¼

b) ½

c) ⅓

d) ⅛

Answer: _____

B. Fill in the Blanks

1. The sum of probabilities of all possible outcomes of an event is _____.

Answer: _____

2. If an event is certain to happen, its probability is _____.

Answer: _____

3. If an event will never happen, its probability is _____.

Answer: _____

4. Probability can never be less than _____ or greater than _____.

Answer: _____

5. The probability of getting an odd number on a dice is _____.

Answer: _____

C. True or False

1. Probability of getting a tail when tossing a coin is ½.

Answer: _____

2. Probability of getting a number less than 7 on a die is 1.

Answer: _____

3. An impossible event has a probability greater than 1.

Answer: _____

4. Probability of all outcomes together is always equal to 1.

Answer: _____

5. Getting “Monday” when a day is chosen randomly from a week has a probability of 1/7.

Answer: _____

D. Short Answer Questions

1. A coin is tossed once. Write the sample space and find the probability of:

(a) Getting a head

(b) Getting a tail

2. A die is rolled. Find the probability of getting:

(a) An odd number

(b) A number less than 4

3. What is the probability of drawing a red card from a standard deck of 52 cards?

4. A box contains 3 red balls and 2 blue balls. If one ball is drawn at random, what is the probability of drawing:

(a) A red ball

(b) A blue ball

5. When a coin is tossed twice, list all possible outcomes. Find the probability of getting:

(a) Two heads

(b) One head

(c) No head

E. Higher Order / Reasoning Questions

1. A bag contains 5 white balls and 3 black balls. One ball is taken out at random.

Find the probability of drawing:

(a) A white ball

(b) A black ball

2. A number is chosen randomly from 1 to 10. Find the probability that it is:

(a) An even number

(b) A multiple of 3

(c) Greater than 5

3. A student is selected at random from a class of 20 students. If 8 are girls, find the probability that the student chosen is:

(a) A girl

(b) A boy

4. A bag contains 10 marbles numbered 1 to 10. If one marble is picked at random, find the probability that it shows:

(a) An even number

(b) A prime number

(c) A number greater than 7

F. Scenario-Based / Real-Life Questions (AI Context)

1. Weather Forecasting:

Google Weather predicts there is a 60% chance of rain tomorrow .

What is the probability that it will not rain ?

Answer: _____

2. Sports Analytics:

A cricket player scores 50 or more runs in 7 out of 10 matches.

What is the probability that he scores 50 or more runs in the next match?

Answer: _____

3. Traffic Prediction:

AI predicts that there is a 0.3 probability of traffic jam in the morning.

What is the probability that there will not be a traffic jam?

Answer: _____

4. Health Prediction:

An AI model predicts a 0.8 probability that a patient will recover from flu using a new medicine.

What is the probability that the patient does not recover ?

Answer: _____

5. AI Coin Experiment:

An AI bot tosses a fair coin 3 times.

What is the probability of getting at least one head?

Answer: _____

Answers

A. Multiple Choice Questions (MCQs)

1. When a coin is tossed once, what is the probability of getting a head?

Answer: ½

2. A die is rolled. What is the probability of getting an even number?

Answer: 3/6 = ½

3. What is the probability of getting a number greater than 4 on a die?

Answer: 2/6 = 1/3

4. If you pick a card at random from a deck of 52, what is the probability of getting a King?

Answer: 4/52 = 1/13

5. Which of the following has a probability of 0?

Answer: Getting 8 on a standard die

6. What is the probability of getting tails twice in two tosses of a fair coin?

Answer: ¼

B. Fill in the Blanks (with Answers)

1. The sum of probabilities of all possible outcomes of an event is 1 .
2. If an event is certain to happen, its probability is 1 .
3. If an event will never happen, its probability is 0 .
4. Probability can never be less than 0 or greater than 1 .
5. The probability of getting an odd number on a die is 3/6 = ½ .

C. True / False (with Answers)

1. Probability of getting a tail when tossing a coin is ½. — True

2. Probability of getting a number less than 7 on a die is 1. — True

3. An impossible event has a probability greater than 1. — False

4. Probability of all outcomes together is always equal to 1. — True

5. Getting “Monday” when a day is chosen randomly from a week has a probability of 1/7. — True

---

D. Short Answer Questions (with Answers)

1. A coin is tossed once. Write the sample space and find the probability of:

(a) Getting a head

(b) Getting a tail

Answer:

Sample space = {H, T}

P(Head) = 1/2, P(Tail) = 1/2

---

2. A die is rolled. Find the probability of getting:

(a) An odd number

(b) A number less than 4

Answer:

Sample space = {1, 2, 3, 4, 5, 6}

(a) Odd = {1, 3, 5} → P = 3/6 = ½

(b) Less than 4 = {1, 2, 3} → P = 3/6 = ½

---

3. What is the probability of drawing a red card from a standard deck of 52 cards?

Answer: 26 red cards in deck → P = 26/52 = ½

---

4. A box contains 3 red balls and 2 blue balls. If one ball is drawn at random, what is the probability of drawing:

(a) A red ball (b) A blue ball

Answer:

Total balls = 5

(a) P(red) = 3/5

(b) P(blue) = 2/5

---

5. When a coin is tossed twice, list all possible outcomes. Find the probability of getting:

(a) Two heads (b) One head (c) No head

Answer:

Sample space = {HH, HT, TH, TT}

(a) Two heads → 1/4

(b) One head → 2/4 = ½

(c) No head → 1/4

E. Higher Order / Reasoning Questions (with Answers)

1. A bag contains 5 white balls and 3 black balls. One ball is taken out. Find P(white) and P(black).

Answer:

Total = 8 balls

P(white) = 5/8 P(black) = 3/8

---

2. A number is chosen randomly from 1 to 10. Find the probability that it is:

(a) An even number (b) A multiple of 3 (c) Greater than 5

Answer:

Total = 10 numbers

(a) Even: {2,4,6,8,10} → 5/10 = ½

(b) Multiples of 3: {3,6,9} → 3/10

(c) Greater than 5: {6,7,8,9,10} → 5/10 = ½

---

3. A student is selected from a class of 20 students. If 8 are girls, find probability of choosing:

(a) A girl (b) A boy

Answer:

(a) 8/20 = 2/5 (b) 12/20 = 3/5

---

4. A bag has 10 marbles numbered 1–10. If one marble is picked, find P(it shows):

(a) Even number (b) Prime number (c) Number > 7

Answer:

(a) Even = {2,4,6,8,10} → 5/10 = ½

(b) Prime = {2,3,5,7} → 4/10 = 2/5

(c) >7 = {8,9,10} → 3/10

F. Scenario-Based / Real-Life Questions (with Answers)

1. Weather Forecasting:

Google Weather predicts 60% chance of rain tomorrow.

What is the probability that it will not rain ?

Answer: P(rain) = 0.6 → P(not rain) = 1 – 0.6 = 0.4

---

2. Sports Analytics:

A cricket player scores 50+ runs in 7 of 10 matches.

What is the probability he scores 50+ in the next match?

Answer: 7/10 = 0.7

---

3. Traffic Prediction:

AI predicts a 0.3 probability of traffic jam in the morning.

What is the probability there is no traffic jam ?

Answer: $1 - 0.3 = 0.7$

---

4. Health Prediction:

An AI model predicts 0.8 probability of recovery from flu.

What is the probability that the patient does not recover ?

Answer: $1 - 0.8 = 0.2$

---

5. AI Coin Experiment:

An AI bot tosses a fair coin 3 times.

Find the probability of getting at least one head .

Answer:

$P(\text{no head}) = (1/2)^3 = 1/8$

$\Rightarrow P(\text{at least one head}) = 1 - 1/8 = 7/8$